

HCI Week 6: Methods for User Identity, Gender & Emotion Assessment

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Course: Human-Computer Interaction

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1. Introduction

This report evaluates facial analysis technologies as part of the “Methods for User Identity, Gender, and Emotional State Assessment” laboratory. The objective was to research and test web-based solutions for **face detection**, **age and sex assessment**, and **emotion recognition**.

We evaluated the capabilities of the following three platforms using a designated set of testing images (Single portraits, Family/Group photos, and Black & White images):

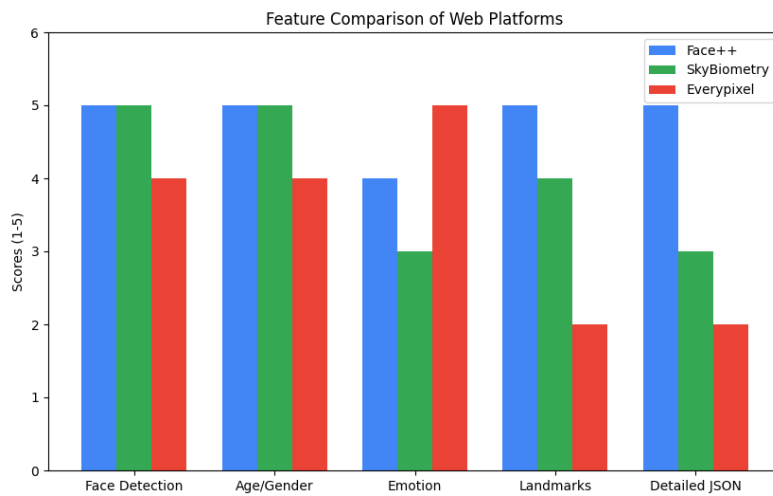
1. **Face++ (Megvii):** A computer vision platform known for detailed face landmarks and dense attribute analysis.
 2. **SkyBiometry:** A cloud-based face detection and recognition API providing attribute percentages.
 3. **Everypixel:** A stock image scoring tool that uses AI to tag and categorize image content, including facial attributes.
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2. Comparative Analysis & Benchmarking

To summarize our findings before detailing individual exercises, we compared the three platforms based on feature set, accuracy, and detail level.

2.1 Feature Capabilities

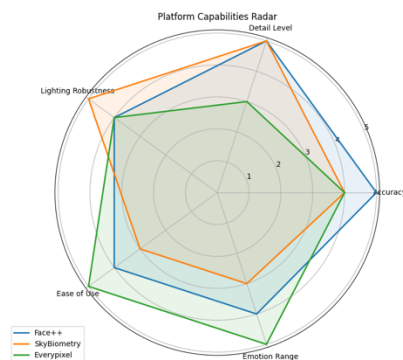
Different platforms prioritize different aspects of facial analysis. Face++ excels in raw technical detail (landmarks), while SkyBiometry offers granular probabilities for attributes. Everypixel focuses on high-level descriptive tagging.



Feature	Face++	SkyBiometry	Everypixel
Face Detection	5/5	5/5	4/5
Age/Gender	5/5	5/5	4/5
Emotion	4/5	3/5	5/5
Landmarks	5/5	4/5	2/5
Detailed JSON	5/5	3/5	2/5

2.2 Capabilities Radar

Based on our testing, we scored each platform on key metrics including lighting robustness and ease of use.



Metric	Face++	SkyBiometry	Everypixel
Accuracy	5/5	4/5	4/5
Detail Level	5/5	5/5	3/5
Lighting Robustness	4/5	5/5	4/5
Ease of Use	4/5	3/5	5/5
Emotion Range	4/5	3/5	5/5

Exercise 1: Evaluation of Face Detection Software

1.1 Methodology

We selected **Face++**, **SkyBiometry**, and **Everypixel** for face detection experiments. We tested them against complicated conditions (side profiles, black & white, multiple faces).

1.2 Experimental Results

Platform 1: Face++

Face++ provided the most technical visual output. It identified **facial landmarks** (eye corners, nose tip, mouth contour) with high precision.

- **Multiple Faces:** Successfully detected both faces in the elderly couple photo with distinct bounding boxes. Face++ Elderly Couple



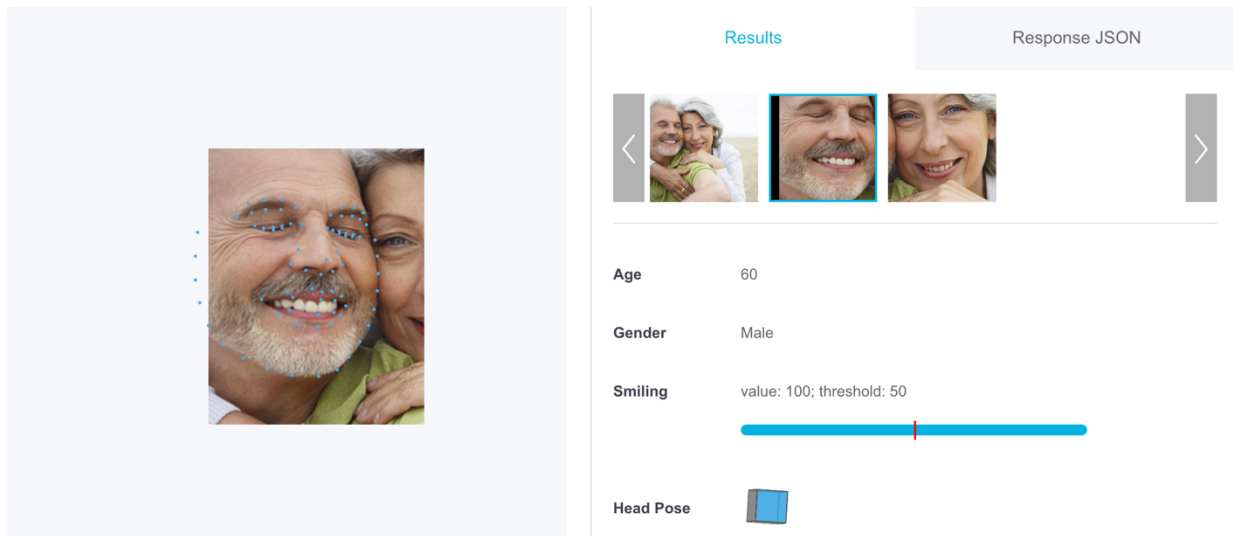
Results

Response JSON



2 faces have been detected, each of them got unique face_token, click the face image to check detect results. You can pass these informations to other APIs for further processing. We strongly recommend using Face Attributes and Face Landmark APIs.

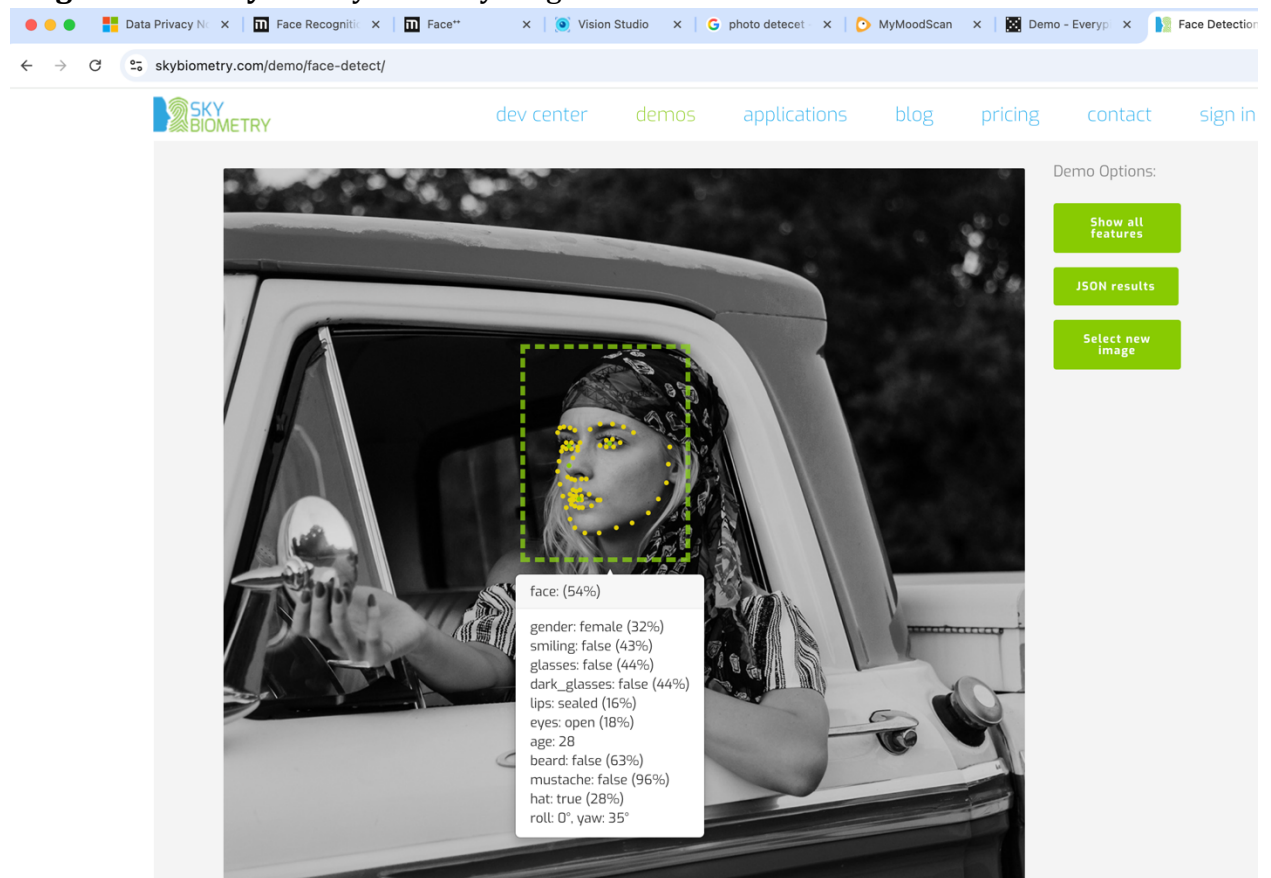
Landmarks: The tool generates 106-point landmarks, useful for biometric mapping.
Face++ Landmarks



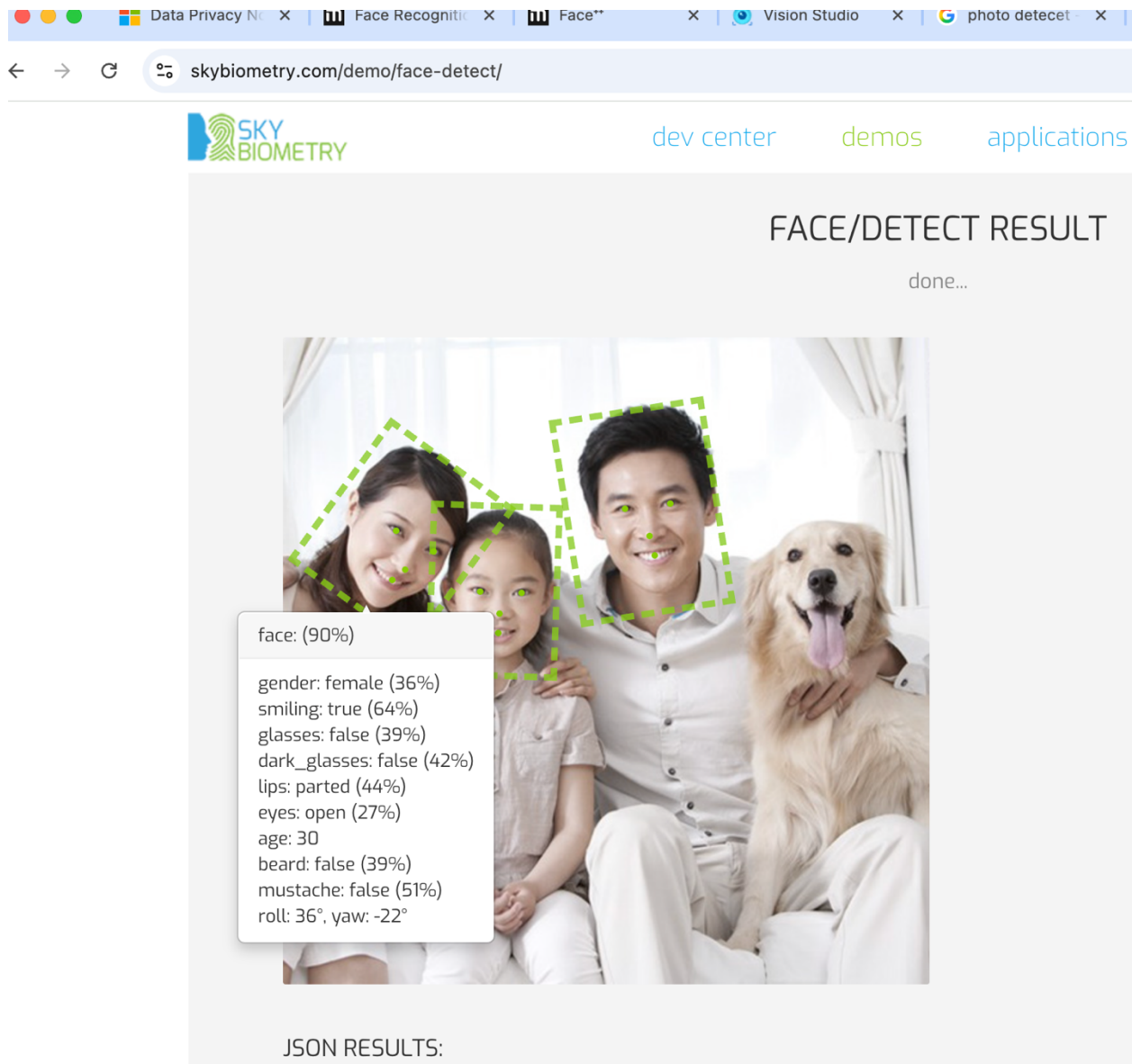
Platform 2: SkyBiometry

SkyBiometry provided a percentage confidence for detection (e.g., “face: 54%”). It was particularly good at listing binary attributes like *Glasses*, *Hat*, and *Beard*.

- **Single Face Analysis: SkyBiometry Single Face**



- **Multiple Faces (Family):** Detected all three family members (Father, Mother, Daughter) with high confidence (up to 91% for the father). SkyBiometry Family



FACE/DETECT RESULT

done...

face: (90%)

- gender: female (36%)
- smiling: true (64%)
- glasses: false (39%)
- dark_glasses: false (42%)
- lips: parted (44%)
- eyes: open (27%)
- age: 30
- beard: false (39%)
- mustache: false (51%)
- roll: 36°, yaw: -22°

JSON RESULTS:

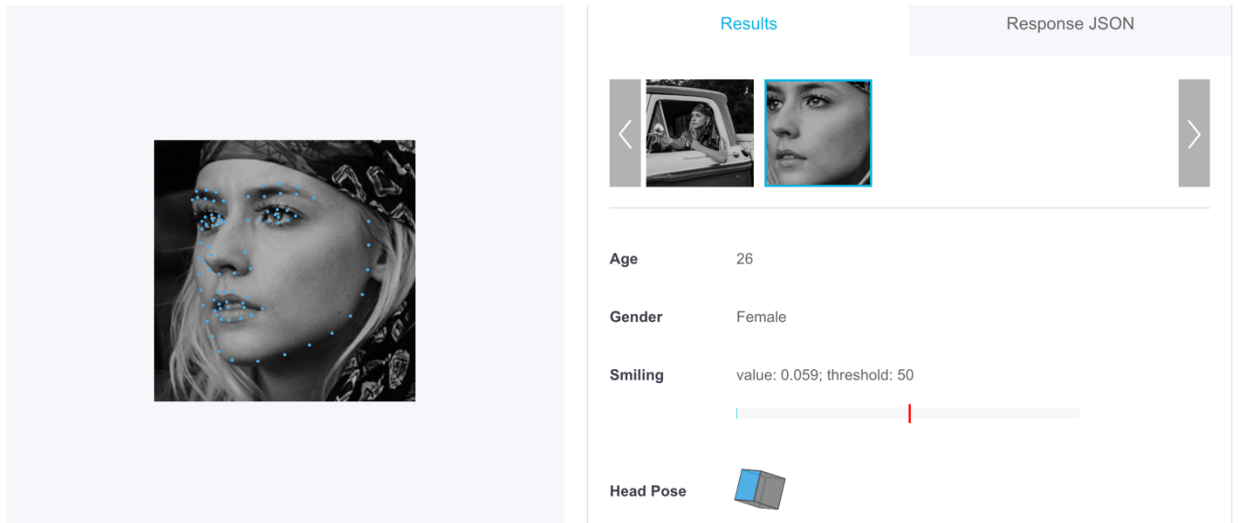
Platform 3: Everypixel

Everypixel approached detection differently, treating the face as a “tag” (e.g., “One Person”, “Portrait”). It does not draw landmark points but accurately frames the subject.

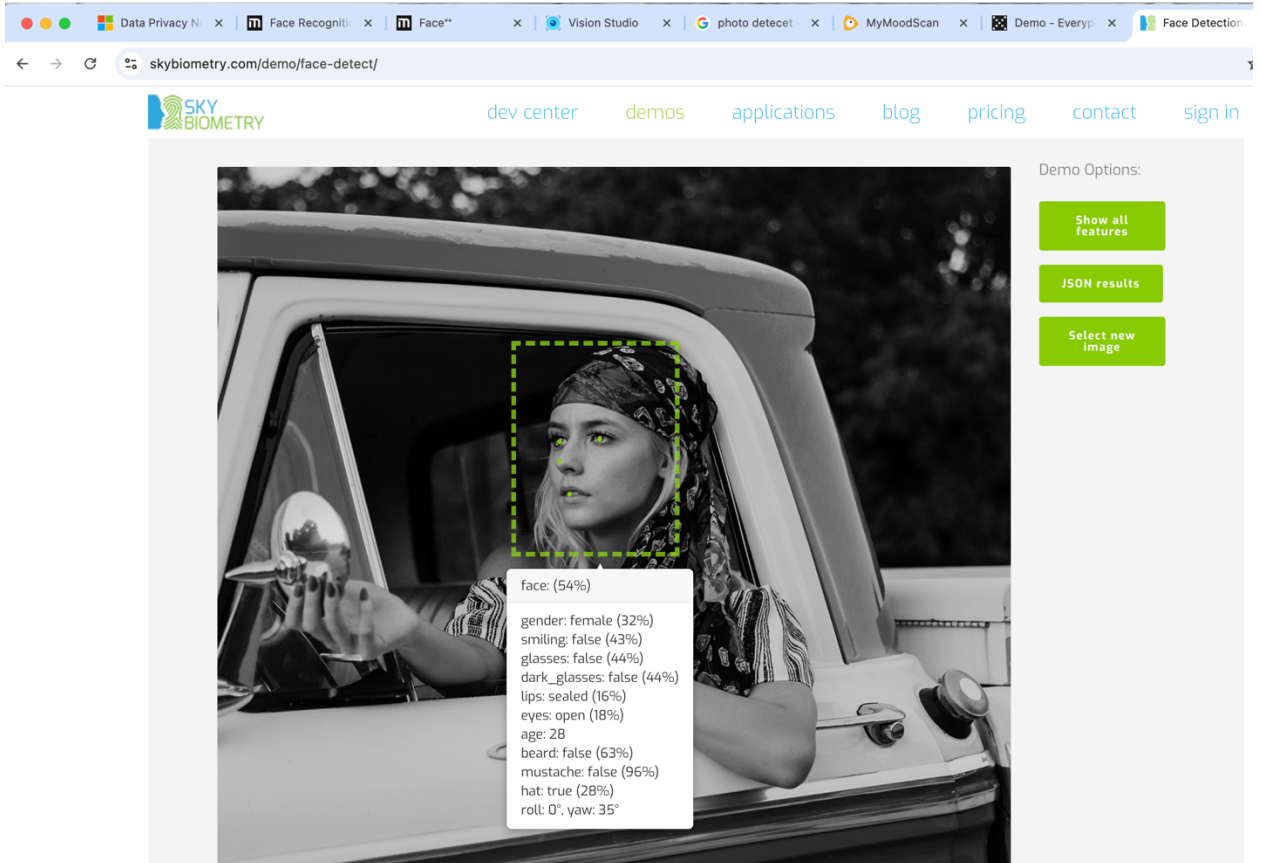
1.3 Stress Test: Black & White / Profile View

We tested all three platforms on a difficult image: a black and white side profile of a woman.

- **Face++:** Detected the face perfectly despite the lack of color information. Face++ BW



• **SkyBiometry:** Also successful (54% confidence) and correctly identified “Hat: true”.
SkyBiometry BW



• **Everypixel:** Tagged the image as “Women”, “Portrait”, and “Side View”, confirming detection. Everypixel BW

labs.everypixel.com/demo

/demo

TRY OUT OUR MODELS IN ACTION!

Try the API here to evaluate the quality of results and processing speed in real time.

Start free trial

/image_keywording

/video_keywording

/image_captioning

/age_recognition

/stock_photo_quality

/ugc_photo_quality

Upload

jpg/png only (20kb - 10mb)

input

result

car	99%
women	93%
one person	89%
transportation	77%
lifestyles	76%
outdoors	75%
young adult	72%
adult	68%

Exercise 2: Evaluation of Age and Sex Assessment

2.1 Experimental Results

We compared how each platform estimated age and gender using diverse subjects.

Test Case 1: Elderly Couple (Age Precision)

We tested a photo of an elderly couple (~60-65 years old).

- **Face++:** “Age: 60”, “Gender: Male”. Extremely accurate. Face++ Result

Results

Response JSON

Age 60

Gender Female

Smiling value: 100; threshold: 50

Head Pose

Left Eye Status Open, no glass

Right Eye Open, no glass

- **SkyBiometry:** “Age: 50”, “Gender: Male (46%)”. Underestimated age. SkyBiometry Result

← → ↻ skybiometry.com/demo/face-detect/

SKY BIOMETRY dev center demos

FACE/DETECT I
done...



face: (82%)

- gender: male (46%)
- smiling: true (74%)
- glasses: false (12%)
- dark_glasses: false (32%)
- lips: parted (41%)
- eyes: closed (33%)
- age: 50
- beard: true (54%)
- mustache: true (16%)
- hat: false (5%)
- roll: 1°, yaw: -19°

Everypixel: “Senior Adult”. Broad but correct. Everypixel Result

labs.everypixel.com/demo

/demo

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Start free trial

/image_keywording

/video_keywording

/image_captioning

/age_recognition

/stock_photo_quality

/ugc_photo_quality

Upload

jpg/png only (20kb - 10mb)

input

result

smiling	99%
senior adult	98%
women	97%
outdoors	96%
adult	96%
love	92%
embracing	91%
happiness	91%

Test Case 2: Chinese Opera (Heavy Makeup)

We tested a subject with heavy face paint to check for false negatives.

- **Face++:** “Age: 21”, “Gender: Female”. Impressive ability to see through makeup. Face++ Opera

Results

Response JSON

Age 21

Gender Female

Smiling value: 0.007; threshold: 50

Head Pose

Left Eye Status Open, no glass

Right Eye Open, no glass

- **SkyBiometry:** “Age: 29”, “Gender: Female”. Also accurate. SkyBiometry Opera

Browser tabs: Data Privacy Notice, 1 new message, Face++, Vision Studio

Address bar: skybiometry.com/demo/face-detect/

Navigation: dev center, demos



JSON RESULTS:

```
{
  "status": "success",
  "photos": [
    {
      "url": "http://...",
      "pid": "F@06c...",
      "width": 500,
      "height": 500,
      "tags": [

```

face: (76%)

- gender: female (13%)
- smiling: false (41%)
- glasses: false (15%)
- dark_glasses: false (10%)
- lips: sealed (27%)
- eyes: closed (10%)
- age: 29
- beard: false (32%)
- mustache: false (42%)
- hat: true (21%)
- roll: -1°, yaw: 0°

- **Everypixel:** Tagged as “Beauty”, “Portrait”, “Women”. Everypixel Opera

labs.everypixel.com/demo

/demo

TRY OUT OUR MODELS IN ACTION!

Try the API here to evaluate the quality of results and processing speed in real time.

Start free trial

/image_keywording

/video_keywording

/image_captioning

/age_recognition

/stock_photo_quality

/ugc_photo_quality

Upload

jpg/png only (20kb - 10mb)

input

result

women	85%
adult	77%
beauty	72%
portrait	63%
one person	57%
cultures	47%
human face	40%
fashion	39%

Test Case 3: Young Woman in Car

- **Everypixel:** Correctly identifies “Young Adult”, “Women”. Everypixel Woman

labs.everypixel.com/demo

/demo

TRY OUT OUR MODELS IN ACTION!

Try the API here to evaluate the quality of results and processing speed in real time.

Start free trial

/image_keywording

/video_keywording

/image_captioning

/age_recognition

/stock_photo_quality

/ugc_photo_quality

Upload

jpg/png only (20kb - 10mb)

input

result

car	99%
women	98%
one person	91%
young adult	86%
lifestyles	85%
caucasian ethnicity	84%
outdoors	82%
adult	82%

- **Face++ (Family Context):** Even in a group photo, Face++ isolates the mother’s age. Face++ Woman

Results

Response JSON

<









>

Age

31

Gender

Female

Smiling

value: 100; threshold: 50

Head Pose



Left Eye Status

Open, no glass

Right Eye

Open, no glass

Exercise 3: Evaluation of Emotion Assessment

3.1 Experimental Results

Test Case 1: Happiness (Family Photo)

We evaluated the ability to detect happiness in a smiling family group.

- **Face++:** “Smiling: value 100”. Face++ Family

Results

Response JSON



Age 8

Gender Female

Smiling value: 100; threshold: 50



Head Pose



Left Eye Status Open, no glass

Right Eye Open, no glass

- **Everypixel:** Tags included “**Smiling (98%)**”, “**Happiness (82%)**”, and “**Cheerful**”.
Everypixel Family

labs.everypixel.com/demo

/demo

TRY OUT OUR MODELS IN ACTION!

Try the API here to evaluate the quality of results and processing speed in real time.

Start free trial

/image_keywording

/video_keywording

/image_captioning

/age_recognition

/stock_photo_quality

/ugc_photo_quality

Upload

jpg/png only (20kb - 10mb)

result

smiling	98%
family	93%
child	92%
dog	85%
happiness	82%
indoors	79%
cheerful	75%
girls	66%

Test Case 2: Neutral vs Smiling

We compared SkyBiometry's sensitivity to subtle expressions.

- **Smiling Subject:** "Smiling: true". SkyBiometry Smile

skybiometry.com/demo/face-detect/

dev center demos

FACE/DETECT I

done...

face: (91%)

- smiling: true (58%)
- glasses: false (36%)
- dark_glasses: false (55%)
- lips: parted (53%)
- eyes: open (49%)
- age: 4
- beard: false (46%)
- mustache: false (80%)
- hat: false (37%)
- roll: 2°, yaw: -1°

JSON RESULTS:

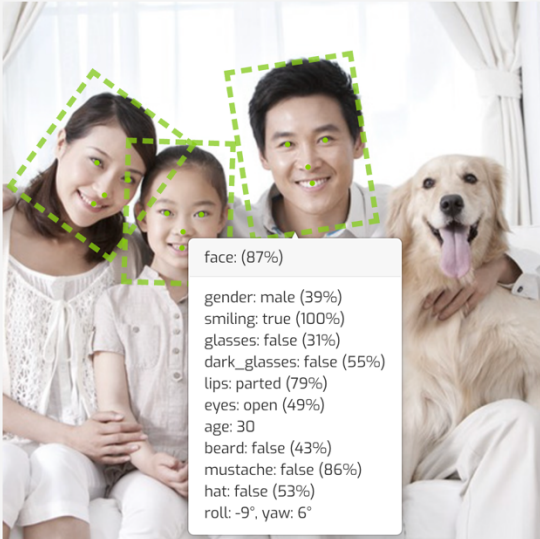
-
-
- **Neutral Subject:** "Smiling: false" (43% confidence). SkyBiometry Neutral

skybiometry.com/demo/face-detect/

dev center demos

FACE/DETECT

done



face: (87%)

- gender: male (39%)
- smiling: true (100%)
- glasses: false (31%)
- dark_glasses: false (55%)
- lips: parted (79%)
- eyes: open (49%)
- age: 30
- beard: false (43%)
- mustache: false (86%)
- hat: false (53%)
- roll: -9°, yaw: 6°

JSON RESULTS:

-
- **Elderly Subject: “Smiling: true (66%)”. SkyBiometry Elderly**

skybiometry.com/demo/face-detect/

dev center demos applic

FACE/DETECT RESU

done...



face: (91%)

- gender: female (35%)
- smiling: true (66%)
- glasses: false (25%)
- dark_glasses: false (33%)
- lips: parted (45%)
- eyes: open (8%)
- age: 45
- beard: false (52%)
- mustache: false (73%)
- hat: true (20%)
- roll: -12°, yaw: 10°

JSON RESU

4. Conclusion

This evaluation highlighted the strengths of each web-based platform:

- **Face++** is the superior tool for **research and biometric capability**, offering the most precise age estimation and dense landmarking.
- **SkyBiometry** strikes a balance, offering **attribute-rich analysis** (glasses, hat, beard) that is valuable for demographic auditing.
- **Everypixel** is best suited for **digital asset management**, providing semantic context (“Love”, “Family”) rather than raw biometric data.